

Algorithms 1–6: Results and Decision Logic

A compact benchmark briefing

medicine-search-clean

4 August 2026

Abstract

The comparison uses 464 fair OCR query–target pairs and 66,257 clean synthetic cases. Algorithm 6 reaches 50.43% Hit@1, 63.58% Hit@5, and 73.28% Hit@20 on OCR. It has the same Hit@1 and Hit@5 as Algorithm 5, but retrieves two additional OCR targets at rank 20. A broader rule that changed rank 1 was rejected because it gained 27 synthetic cases and lost 138.

1 Read the Numbers First

A **medicine family** is the commercial-name group being ranked. A **candidate** is one family shown to the user. A **retrieval source** is one search method that independently returns a ranked candidate list.

Dataset	Cases	Hit@1	Correct	Hit@5	Correct	Hit@20
Fair OCR pairs	464	50.43%	234	63.58%	295	73.28% (340)
Synthetic clean core	66,257	98.19%	65,057	99.65%	66,027	99.9985% (66,256)

Table 1: Algorithm 6 headline counts. Hit@1 means the expected family is first; Hit@5 and Hit@20 mean it appears within the first five or twenty families. The green and blue columns identify Hit@1 and Hit@5 throughout the report.

The OCR set gives one vote to each unique fair query–target pair. The synthetic set applies generated spelling errors to catalog-derived targets and excludes unfair cases where the corrupted query is exactly another real family name.

2 What Changed from Algorithm 1 to Algorithm 6

The algorithm names are their version numbers. The descriptions below explain the change; they are not additional algorithm names.

Algorithm	Change introduced in that version	OCR ms/query
Algorithm 1	Original application search over medicine names.	1.28
Algorithm 2	Independent fast typo search that narrows the catalog before comparison.	4.75
Algorithm 3	Combines the ranked lists produced by Algorithms 1 and 2.	6.84
Algorithm 4	Adds family and visible-fragment recovery for longer product names.	14.89
Algorithm 5	Adds position, missing-letter, sound, and family evidence learned from failure analysis.	28.54
Algorithm 6	Keeps Algorithm 5's order and may add one independently supported missing family at rank 20.	48.50

Table 2: Each version adds a specific capability and increases mean OCR query time. Algorithm 6 does not replace Algorithm 5's first result.

Method	464 fair OCR pairs			66,257 synthetic pairs		
	H@1	H@5	H@20	H@1	H@5	H@20
Algorithm 1	23.71	31.47	38.15	83.86	89.72	92.23
Algorithm 2	25.86	43.75	51.29	92.41	96.33	96.88
Algorithm 3	25.65	40.95	53.02	91.84	96.18	97.12
Algorithm 4	47.63	61.85	71.12	97.19	99.24	99.52
Algorithm 5	50.43	63.58	72.84	98.19	99.65	99.9985
Algorithm 6	50.43	63.58	73.28	98.19	99.65	99.9985

Table 3: Percent recovery. Algorithm 6 improves Algorithm 5 only at Hit@20: 340 rather than 338 OCR pairs.

Method	464 fair OCR pairs			66,257 synthetic pairs		
	H@1	H@5	H@20	H@1	H@5	H@20
SymSpell, edit distance 3	38.58	50.86	55.82	94.22	98.08	98.80
Damerau-Levenshtein	34.27	50.65	59.27	94.78	98.69	99.64
Exhaustive Levenshtein	34.27	50.86	59.27	94.05	98.40	99.44
Jaro-Winkler	32.11	55.17	67.46	89.57	96.26	98.43
BM25+ character trigrams	20.26	35.13	47.41	78.19	87.79	92.17
BioSyn-style hybrid	19.40	39.66	50.86	77.88	88.88	93.40
Character 3-gram TF-IDF	18.32	35.78	47.41	75.21	86.72	91.66
xMEN-style dense+sparse fusion	16.81	37.50	52.37	48.85	78.51	92.30
SapBERT dense retrieval	11.42	22.63	38.58	27.67	44.00	56.45
RapidFuzz weighted ratio	11.21	40.30	56.25	55.73	93.49	97.20
Phonetic baseline	6.90	16.81	23.28	69.66	84.92	92.07
Exact or prefix match	2.59	2.80	3.02	3.00	3.01	3.01

Table 4: Percent recovery for the main non-project method families. The same input, catalog, family-level scoring, and denominator are used for every row.

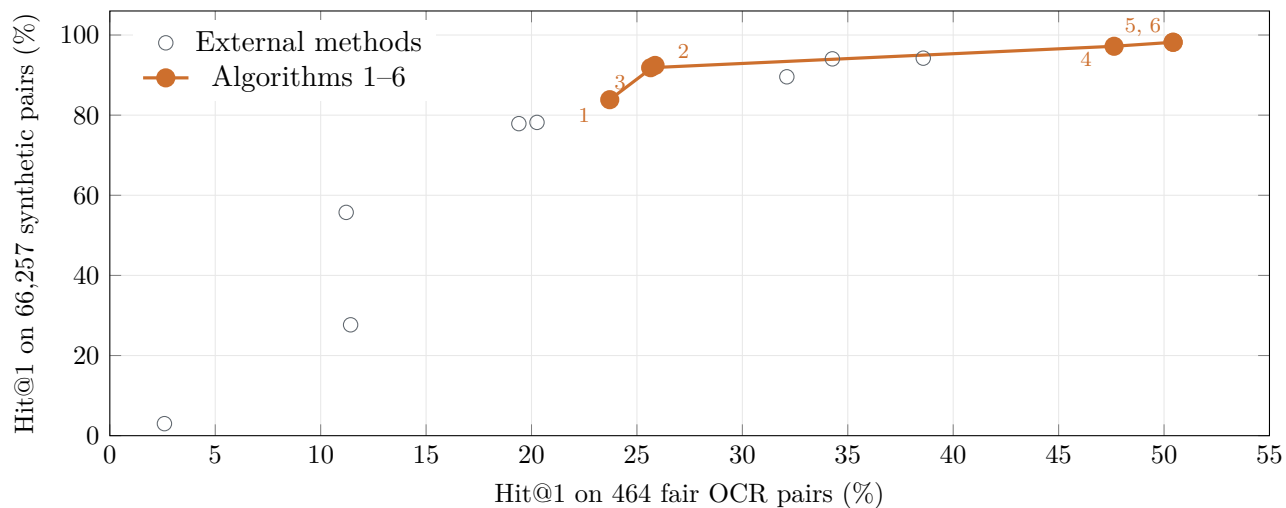


Figure 1: Each point uses OCR Hit@1 on the x-axis and synthetic Hit@1 on the y-axis. Numbers beside orange points identify Algorithms 1–6. Synthetic results are much higher for nearly every method, so the 66,257-case score cannot replace OCR evaluation. Algorithms 5 and 6 occupy the same point because Algorithm 6 does not change rank 1.

3 Where the Remaining Retrieval Problem Starts

Compact edit distance counts the fewest single-character insertions, deletions, or replacements after case, spaces, and punctuation are removed. For example, *TEFA*→*TELF**AST* needs three insertions.

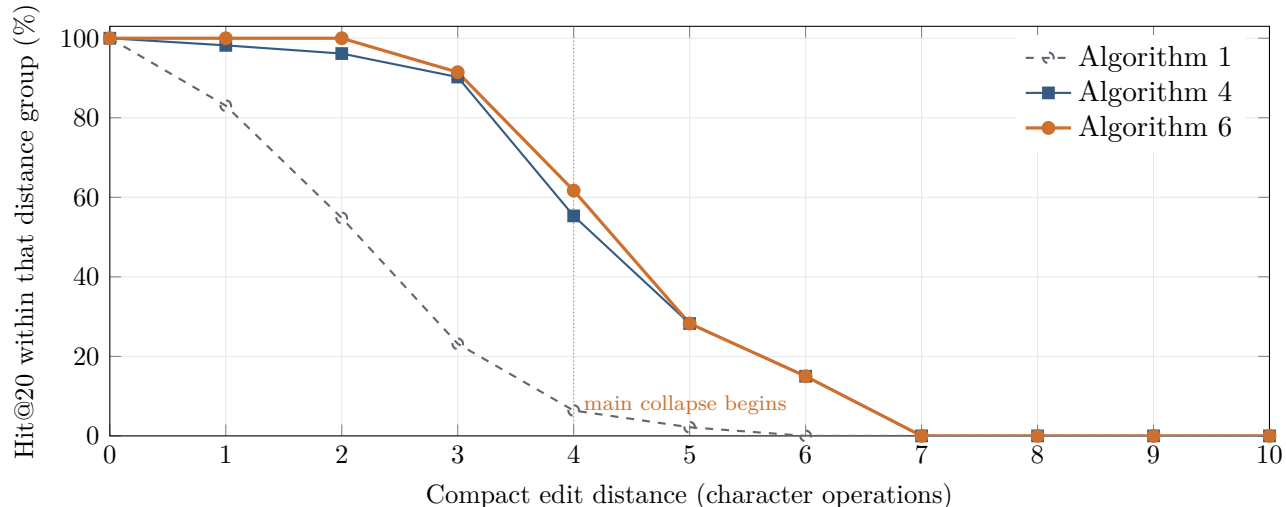


Figure 2: The x-axis is corruption severity and the y-axis is retrieval within the first 20. Algorithm 6 retrieves every one- and two-edit OCR case, 91.46% of three-edit cases, 61.70% at four edits, and none at seven or more. The next retrieval work should target four-to-six-edit evidence rather than easy typos.

4 How Algorithm 6 Makes One Decision

Algorithm 6 starts with Algorithm 5’s top 20. Nine additional methods search the same catalog. Names referring to the same family are merged. A family absent from Algorithm 5’s list can enter only the final slot when all three rules pass:

1. at least three retrieval sources return it;
2. Levenshtein similarity is at least 55%;
3. Jaro–Winkler similarity is at least 85%.

Levenshtein similarity is $1 - d / \max(|q|, |c|)$, where d is edit distance, q is the query, and c is the candidate. Jaro–Winkler is a 0–1 score that rewards matching characters in the same order and gives extra weight to a shared prefix. Algorithm 6 adds at most one family at rank 20 and never changes rank 1.

Algorithm 5 is not one opaque component inside Algorithm 6. It is the complete base pipeline summarized below. Algorithm 6 then adds the independent sources, family-level consensus, bounded insertion gate, and confirmation policy.

Nested Algorithm 5 component	Cases it handles	Concrete benchmark example
Normalization and context cleanup	Case, punctuation, spacing, strength, route, and form text around a name.	AAPE RENEWAL SUNSCREEN SPF 60ML→AAPE RENEWAL SUNSCREEN SPF.
External candidate retriever	Broad typo candidates and a second pass after context cleanup.	Removing it loses ABDUSONG LAR→ABASAGLAR from Algorithm 6’s top 20.
Family-rescue index	Exact, prefix, suffix, character n-gram, consonant, phonetic, delete-key, and compatible-length lookup over 17,476 families.	MYMLAX→MYOLAX survives a middle-character substitution.
String-evidence scores	Raw and weighted edit similarity, prefix, suffix, n-grams, phonetic code, skeleton, subsequence, position, and length coverage.	TEFAXATE→TELFAS uses position and shared-edge evidence.
Error-variant retrieval	Keyboard, ligature, transposition, visual, phonetic, missing-letter, and bounded multi-step transformations.	Synthetic uraranx→ORAMAX depends on the multi-step family in the non-regression set.
Family-head and short-fragment rescue	Matches visible family heads and short prefix/suffix evidence without treating every substring as decisive.	LANS→LANTUS is lost from top 20 when short visible-head rescue is removed.
Deterministic rerankers	Correct equal-distance and evidence-dominance cases after candidate merging.	LANYU→LANTUS falls outside top 20 when all rerankers are removed.
Clarification gate	Keeps the ranking but marks weak, short, or conflicting evidence for user confirmation.	RIV→RIVOTRIL competes with the shorter catalog family RIVO; the system must not certify one answer.

Table 5: The internal Algorithm 5 pipeline that Algorithm 6 executes first. The later ablation removes these components inside Algorithm 6 and then leaves all nine additional sources and the consensus gate active.



Figure 3: Complete decision for TEFA. The hidden benchmark target is TELFAST, but it is never shown to search. Independent lists provide the three votes; the two similarity checks pass; the candidate is added at rank 20. The top result remains TERA, so the interface must request confirmation.

Retrieval source	What it measures	Actual case it retrieves	Measured role
Algorithm 5	Complete base pipeline above	MYMLAX→MYOLAX at rank 1	Preserves all deployed rank-1 ordering.
Jaro–Winkler	Character order and shared prefix	TEFA→TELFAS at source rank 1	Required for the TEFA top-20 gain.
Match Rating	Compressed consonant pattern	ABASCOLOGY→ABASAGLAR at source rank 1	No isolated top- <i>k</i> change on 464 OCR pairs.
RapidFuzz weighted ratio	Best direct or partial string ratio	LACTULOSE→LACTO at source rank 3	Supports LACTO; redundant under one-source removal.
NYSIIS	Pronunciation-oriented code	MYOLOGY→MYOLAX at source rank 1	No isolated top- <i>k</i> change on 464 OCR pairs.
BM25+ trigrams	Weighted shared three-character pieces	LACTULOSE→LACTO at source rank 12	Required to select the LACTO gain over another outside candidate.
SymSpell	Candidates within three delete-key edits	OSFOCU→OSTOCAL at source rank 1	No isolated top- <i>k</i> change on 464 OCR pairs.
RapidFuzz token-sort	Token/string comparison after normalization	TEFA→TELFAS at rank 2; LACTULOSE→LACTO at rank 5	Removing it loses one deployed gain.
Soundex	Coarse first-letter and consonant sound	TELBACE→TELFAS at source rank 1	Broad phonetic backstop; never sufficient alone.
Dice bigrams	Shared two-character pieces	TEFA→TELFAS at source rank 15	Supplies TEFA’s third required vote.

Table 6: Every candidate source is tied to an actual benchmark row. “No isolated change” means the remaining sources preserved the same top-*k* outcome on this sample; it does not mean the source never retrieves useful candidates.

Input	Supplied target	Sources	Lev.	Jaro	A5 → A6 rank	Unchanged top 1
TEFA	TELFAS	3	57.1%	88.6%	> 20 → 20	TERA
LACTULOSE	LACTO	5	55.6%	91.1%	> 20 → 20	LACTULOSE HEK

Table 7: Both rows pass the three-source and two-similarity rules. They are P0 and P1 audit cases, so they prove a retrieval change, not that the supplied target is clinically or visually correct. Source-crop review is still required.

5 Ablation: What Actually Contributes

A true Algorithm 6 ablation removes one component, reruns the changed Algorithm 5 base, and then executes the other nine retrieval sources and the same consensus gate. It does not copy an Algorithm 5-only percentage. A percentage-point change is absolute: 73.276% to 72.845% is -0.431 points. “G/L” reports paired gains and losses relative to complete Algorithm 6.

Nested component removed	H@1	Δ H@1	H@20	Δ H@20	H@20 G/L	Actual affected case
Family rescue	36.638	-13.793	56.034	-17.241	3/83	TUTORIALAE→KETOROLAC: rank 7 to > 20.
All deterministic reranking	40.517	-9.914	70.690	-2.586	0/12	LANYU→LANTUS: rank 2 to > 20.
Weighted edit similarity	49.569	-0.862	66.164	-7.112	2/35	The largest scoring-signal loss at top 20.
Raw edit similarity	49.353	-1.078	67.888	-5.388	2/27	Removes the uniform character-distance score.
Short-edge retrieval	49.784	-0.647	70.259	-3.017	0/14	RICOHIL→RIVOTRIL: rank 3 to > 20.
Length coverage	50.216	-0.216	70.259	-3.017	1/15	Penalizes candidates whose visible-length support is weak.
Family-head rescue	48.060	-2.371	70.905	-2.371	0/11	LANYU→LANTUS: rank 2 to > 20.
Delete-key retrieval	48.276	-2.155	71.121	-2.155	0/10	Recovers names explained by missing characters.
Prefix evidence	48.707	-1.724	71.336	-1.940	0/9	RIVITAM→RIVOTRIL: rank 17 to > 20.
Positional evidence	48.922	-1.509	71.336	-1.940	1/10	TEFAXATE→TELFAX: rank 16 to > 20.
Character n-grams	48.060	-2.371	71.983	-1.293	2/8	Local character pieces retain evidence around OCR errors.
External retriever	49.784	-0.647	72.414	-0.862	0/4	ABDUSONG LAR→ABASAGLAR: rank 1 to > 20.
Weighted confusion costs	48.707	-1.724	72.845	-0.431	3/5	KETOMAN→KATOMOR: rank 14 to > 20.
Phonetic evidence	49.569	-0.862	73.276	0.000	0/0	ECOFIXIM→CEFIXIME: rank 1 falls to rank 2.
Short visible-head rescue	50.431	0.000	73.060	-0.216	0/1	LANS→LANTUS: rank 20 to > 20.

Table 8: End-to-end OCR ablations over the same 464 pairs. Algorithm 6 remains active after each nested removal, so its independent sources may recover cases lost by the modified base. Gains and losses are not additive because several components can affect the same row.

The complete CSV contains 96 nested removals, including atomic flags. Two cross-dataset results change the interpretation. Removing all multi-step chains has zero OCR top- k effect because other sources cover the same OCR rows, but loses 343 synthetic Hit@1 cases while gaining 3, and loses 178 synthetic Hit@20 cases with no gains. Removing family-head rescue loses 11 OCR Hit@20 cases but gains 22 net synthetic Hit@1 cases and recovers the single synthetic top-20 miss, `aclox`→`ADOX`. That conflict needs a narrower guard, not deletion of the whole component.

Algorithm 6 component removed	OCR H@20	Cases lost	Δ H@20	Why
Bounded rank-20 slot	72.845%	2	−0.431 pp	Both TEFA and LACTULOSE return outside the top 20.
Jaro-Winkler	73.060%	1	−0.216 pp	TEFAST falls below three supporting sources.
RapidFuzz token-sort	73.060%	1	−0.216 pp	The same TEFA case loses one required vote.
Dice bigrams	73.060%	1	−0.216 pp	Shared pieces TE and FA supplied the third vote.
BM25+ trigrams	73.060%	1	−0.216 pp	A competing outside candidate takes the only slot instead of LACTO.
Match Rating, WRatio, NYSIS, SymSpell, or Soundex, one at a time	73.276%	0	0.000 pp	Other sources preserve both selected candidates on this sample.

Table 9: Ablation of the Algorithm 6-specific consensus layer. These rows are separate from Table 8, which opens the Algorithm 5 base and removes its internal components.

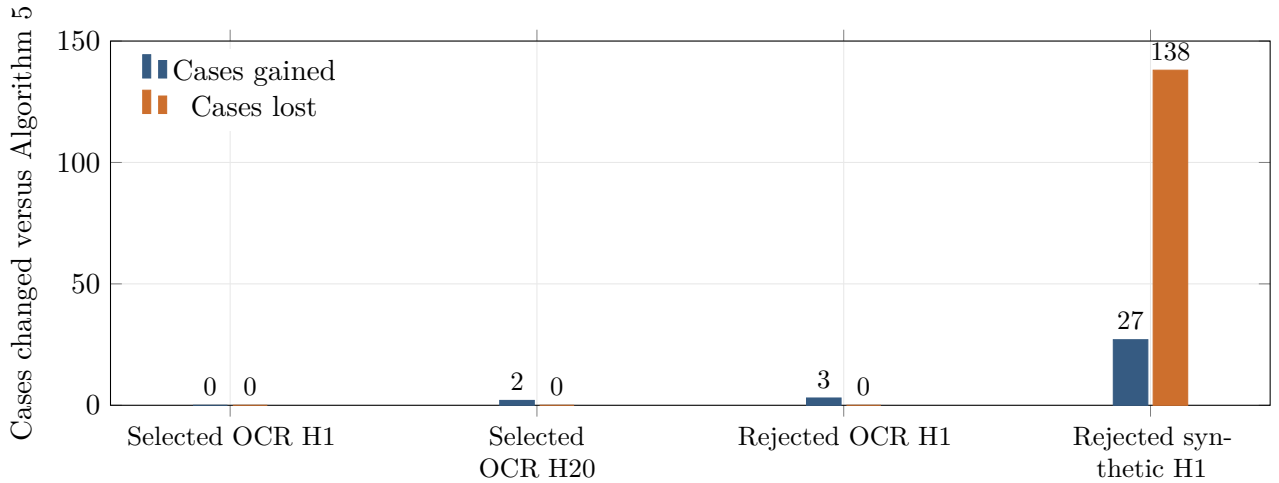


Figure 4: The selected policy changes only OCR Hit@20 and gains two cases with zero losses. The rejected rule would change rank 1: it gains three OCR cases, but on the 66,257 synthetic pairs it gains 27 and loses 138. This transfer failure is why Algorithm 6 preserves Algorithm 5’s first result.

Tested alternative	Measured result	Decision
Allow consensus to change rank 1	OCR H@1 gains 3 and loses 0, but synthetic H@1 gains 27 and loses 138. Net synthetic change is −111 cases, or −0.168 percentage points.	Reject
Replace deterministic order with learned reranking	On the 93-family-disjoint holdout pairs, Hit@1 falls from 48.39% to 46.24% and Hit@20 falls from 79.57% to 77.42%.	Reject
Bounded rank-20 consensus	OCR Hit@20 gains 2 and loses 0; OCR Hit@1, Hit@5, and all synthetic top-k metrics remain unchanged.	Keep for evaluation

Table 10: Only the bounded rank-20 change survives both OCR and synthetic paired checks. It remains an experimental benchmark policy, not a fresh clinical validation.

6 Failure Audit and Work Order

P0, P1, and P2 are automated review priorities, not human verdicts. P0 means a strong label or visible-evidence conflict. P1 means another catalog family is one edit closer. P2 means no automated conflict was found and the supplied target appears in the measured candidate union.

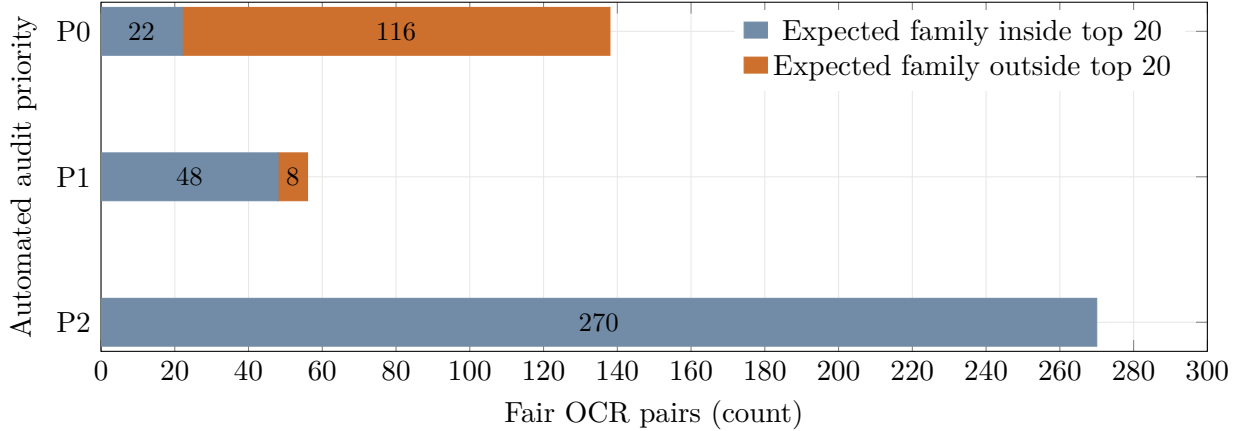


Figure 5: All 124 Algorithm 6 top-20 misses are in the review queues: 116 of 138 P0 pairs and 8 of 56 P1 pairs. P2 has 270 retrieved pairs and zero misses, but target presence in the candidate union is part of the P2 rule; this is not an independent 100% accuracy claim.

Audit	Pairs	Example	Meaning and action
P0	138	CINNARIZIN→MYOLAX; catalog CINNARIZINE is one edit away.	Strong conflict. Check the original crop and target before changing search or using the row for learning.
P1	56	ABASCOLOGY→ABASAGLAR; SPASCOLON is one edit closer.	Borderline conflict. Keep in diagnostic metrics and send for review.
P2	270	MYMLAX→MYOLAX; the supplied family is retrieved.	Provisional clean evidence. It is eligible for cross-fitted analysis, not declared human-verified truth.

Table 11: Every audit class is grounded by one row. None of the 464 pairs has a linked source-image reference, so automated priority cannot replace adjudication.

Among the 124 top-20 misses, 97 supplied targets are absent from all ten source lists. Twenty-six targets appear in only one or two lists and fail at least one similarity rule. One target appears in four lists but fails the 55% Levenshtein rule. No missed target passes every gate and then loses the single external slot; the remaining problem is missing or weak evidence, not slot capacity.

OCR input	Supplied target	A6 top 1	Expected-family evidence	Why it is outside top 20, and what the row shows
CINNARIZIN	MYOLAX	CINNARIZINE	0 sources; target distance 9; nearest distance 1	Candidate-generation miss caused by a likely label conflict. Tuning toward MYOLAX would contradict the visible text.
H	RIVO	no result	0 sources; one visible character	There is not enough evidence to identify one family. Returning no result is safer than manufacturing similarity.
HOAVACH	AMIKACIN	HYABAK	0 sources; target distance 6; nearest distance 3	Severe mixed corruption removes the target from every selected list. This needs stronger retrieval or crop review, not reranking.
LANSOPRAZOLE	LANTUS	PANTOPRAZOLE	1 source; Lev. 33.3%; Jaro 81.1%	It fails all three gate requirements. The returned family is two edits away while the supplied target is nine edits away.
OSCAR	OSTOCAL	ASCOR	2 sources; Lev. 57.1%; Jaro 83.2%	Levenshtein passes, but source count and Jaro fail. This is a borderline P1 row where lowering one threshold is insufficient.
RIV	RIVOTRIL	RIVO	2 sources; Lev. 37.5%; Jaro 85.4%	The prefix supports several families and an exact short family is closer. More visible letters or user confirmation are required.
NEUROTROPHIC	NEUROCET	NEUROTOP	4 sources; Lev. 41.7%; Jaro 85.0%	Consensus and Jaro pass; only Levenshtein blocks insertion. The rule prevents four weak votes from overriding poor character agreement.
ABACAVIR	ABASAGLAR	TECAVIR	1 Soundex source; Lev. 31.6%; Jaro 65.3%	A phonetic source alone cannot justify insertion, and TECAVIR is one edit closer than the supplied family.
LANLTA	LANTUS	LANETAL	2 sources; Lev. 33.3%; Jaro 80.6%	The supplied family is found weakly at source ranks 18 and 20; LANETAL is one edit closer.
TOCILIZUMAB	TOCO	ALIZOLAM	1 WRatio source; Lev. 27.3%; Jaro 67.4%	The four-letter supplied target discards most observed characters. Source-image adjudication must precede any algorithm change.

Table 12: Ten representative Algorithm 6 top-20 failures. “Sources” counts the ten ranked lists containing the supplied family. The insertion gate needs at least three sources, Levenshtein similarity at least 55%, and Jaro–Winkler at least 85%. P0 and P1 remain automated review priorities, not label verdicts.

Order	Work item	Why it comes here
1	Link source crops and adjudicate P0/P1 rows.	Every remaining top-20 miss is P0 or P1. Improving agreement with an incorrect supplied target would make the system worse.
2	Keep Algorithm 5’s rank 1 fixed.	The tested rank-1 rule loses 111 net synthetic cases. No measured evidence supports deploying it.
3	Keep the rank-20 slot, Jaro–Winkler, BM25+, token-sort, and Dice.	These produce the only two accepted gains. Test removal of zero-delta sources on fresh reviewed cases before simplifying them.
4	Reduce Algorithm 6 latency without changing output.	Mean OCR time is 48.50 ms versus 28.54 ms for Algorithm 5. Cache or parallelize independent sources and require exact rank parity.
5	Run the pharmacist within-subject study.	Compare no tool, DrugEye, and Algorithm 6 on the same reviewed crops; allow “cannot decide” as a correct safe action.

Table 13: The order protects label integrity first, then ranking safety, component evidence, runtime, and finally human-use validation.